

Fuzzy Logic-Driven Adaptive Semantic Communication for the Internet of Vehicles

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Abstract—Semantic communication has emerged as a promising solution for efficient data transmission in Internet of Vehicles (IoV) environments. However, conventional semantic communication systems either transmit all data through semantic encoders regardless of channel conditions or rely on direct transmission strategies that fail to adapt to dynamic vehicular environments. In this paper, we propose a Fuzzy Logic-Driven Adaptive Semantic Communication framework for IoV that dynamically selects transmission strategies based on real-time channel quality, distance, and vehicle mobility conditions. The proposed framework integrates fuzzy logic decision-making with adaptive semantic transmission to improve communication quality while reducing unnecessary computational overhead. Experimental results demonstrate that the proposed method significantly improves Peak Signal-to-Noise Ratio (PSNR) and transmission accuracy compared with direct transmission approaches, while consuming fewer computational resources than fully semantic transmission schemes. Extensive evaluations under varying signal-to-noise ratio (SNR), distance, and relative speed conditions validate the effectiveness and robustness of the proposed approach for intelligent vehicular communications.

Index Terms—Semantic communication, Internet of Vehicles (IoV), fuzzy logic, adaptive transmission, vehicular networks, PSNR, intelligent transportation systems.

I. INTRODUCTION

THE rapid development of intelligent transportation systems and autonomous driving technologies has significantly increased the demand for reliable and efficient vehicular communications. Internet of Vehicles (IoV) applications require ultra-reliable and low-latency communication under highly dynamic wireless environments characterized by rapidly changing channel conditions, vehicle mobility, and interference.

Traditional communication systems based on Shannon theory focus on accurate bit-level transmission, often leading to inefficient resource utilization in bandwidth-constrained vehicular environments. Recently, semantic communication has emerged as a promising paradigm that prioritizes the transmission of semantic information rather than exact bit reconstruction. By leveraging deep learning-based semantic encoders and decoders, semantic communication systems can achieve higher robustness and compression efficiency under noisy channels.

Despite these advantages, existing semantic communication approaches face two major limitations in IoV scenarios:

- Always transmitting through semantic models introduces unnecessary computational overhead under favorable channel conditions.

- Direct transmission approaches fail to maintain communication quality under severe channel degradation and mobility.

To address these challenges, this paper proposes a fuzzy logic-driven adaptive semantic communication framework that dynamically switches between transmission strategies according to channel conditions and vehicular dynamics.

The main contributions of this paper are summarized as follows:

- We propose a fuzzy logic-based adaptive transmission framework for semantic communication in IoV environments.
- We design a dynamic decision-making mechanism that jointly considers SNR, transmission distance, and relative vehicle speed.
- We develop an adaptive semantic communication architecture that balances communication quality and computational efficiency.
- We conduct extensive experiments under varying vehicular conditions and demonstrate that the proposed method improves PSNR and transmission accuracy compared with direct transmission methods while reducing computational costs compared with fully semantic transmission schemes.

The remainder of this paper is organized as follows. Section II reviews related works. Section III introduces the proposed system model and fuzzy logic framework. Section IV presents the experimental setup. Section V discusses the experimental results and analysis. Finally, Section VI concludes the paper and outlines future research directions.

II. RELATED WORK

A. Semantic Communication Systems

- Overview of semantic communication theory
- Deep learning-based semantic encoding and decoding
- Joint source-channel coding (JSCC)
- Semantic communication for wireless networks

B. Semantic Communication in Vehicular Networks

- Challenges in IoV communications
- Dynamic channel characteristics in vehicular environments
- Existing semantic communication approaches for IoV

C. Adaptive Transmission Techniques

- Adaptive modulation and coding
- SNR-aware transmission strategies
- Resource allocation in vehicular communications

D. Fuzzy Logic in Wireless Communications

- Fundamentals of fuzzy logic systems
- Fuzzy decision-making for network optimization
- Applications in intelligent vehicular systems

III. SYSTEM MODEL AND PROPOSED METHOD

A. Overall System Architecture

- Transmitter architecture
- Receiver architecture
- Semantic encoder-decoder pipeline
- Wireless channel model

B. Vehicular Communication Scenario

- IoV network assumptions
- Mobility model
- Distance and relative speed considerations

C. Adaptive Semantic Communication Framework

- Direct transmission mode
- Semantic transmission mode
- Hybrid adaptive switching mechanism

D. Fuzzy Logic Decision System

1) Input Variables:

- Signal-to-Noise Ratio (SNR)
- Vehicle distance
- Relative speed

2) Membership Functions:

- Low/Medium/High SNR
- Near/Mid/Far distance
- Slow/Moderate/Fast relative speed

3) Fuzzy Rules: Example rules:

- IF SNR is High AND Distance is Near THEN Direct Transmission
- IF SNR is Low OR Distance is Far THEN Semantic Transmission
- IF Relative Speed is High THEN Adaptive Semantic Mode

4) Defuzzification Strategy:

- Decision score computation
- Final transmission mode selection

E. Complexity Analysis

- Computational complexity
- Resource consumption analysis
- Comparison with fully semantic transmission

IV. EXPERIMENTAL SETUP

A. Dataset Description

- CIFAR-10 dataset
- Data preprocessing
- Training-validation split

B. Neural Network Architecture

- Semantic encoder architecture
- Semantic decoder architecture
- Classifier model

C. Wireless Channel Configuration

- AWGN channel
- Rayleigh fading model (optional)
- SNR range settings

D. Evaluation Metrics

- Peak Signal-to-Noise Ratio (PSNR)
- Classification accuracy
- Computational complexity
- Resource utilization
- Transmission latency

E. Baseline Methods

- Direct transmission
- Fully semantic transmission
- Fixed-SNR semantic transmission
- Proposed fuzzy logic adaptive method

V. EXPERIMENTAL RESULTS AND DISCUSSION

A. PSNR Performance Analysis

- PSNR versus SNR
- PSNR versus transmission distance
- PSNR versus relative vehicle speed

B. Transmission Accuracy Analysis

- Classification accuracy comparison
- Robustness under noisy channels
- Performance under mobility

C. Computational Resource Analysis

- Encoder activation frequency
- Computational savings
- Trade-off between quality and complexity

D. Adaptive Decision Behavior

- Fuzzy decision visualization
- Switching behavior analysis
- Interpretation of adaptive transmission choices

E. Discussion

- Advantages of fuzzy logic adaptation
- Impact on IoV reliability
- Scalability considerations
- Limitations of the current framework

Transmission Performance Comparison (Binned Analysis)

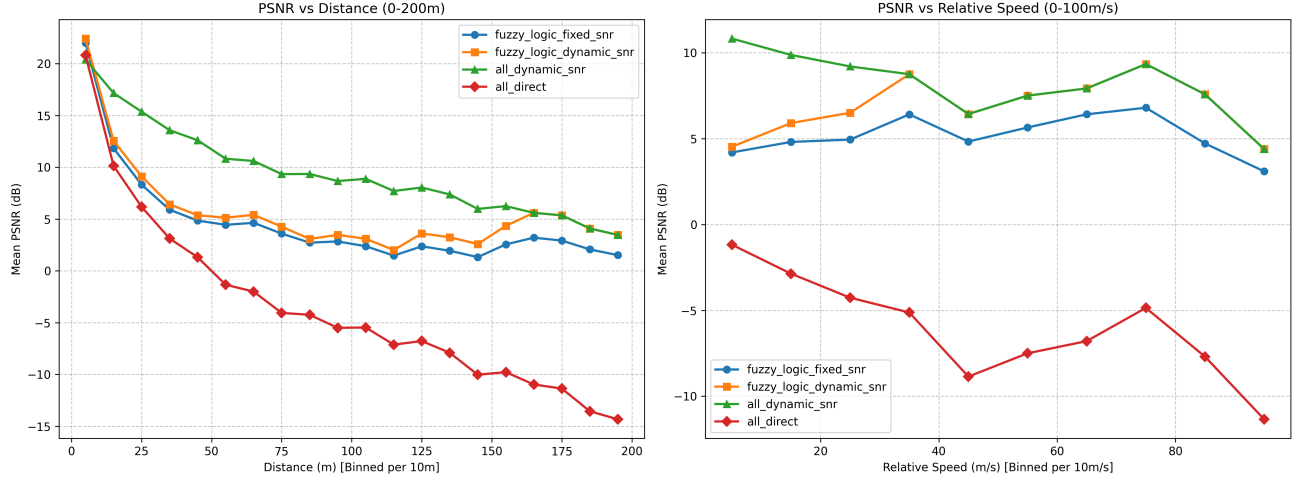


Fig. 1. Transmission performance comparison under varying distance and relative speed conditions. The proposed fuzzy logic-driven adaptive semantic communication method achieves higher PSNR performance than direct transmission while maintaining lower computational overhead than fully semantic transmission.

VI. CONCLUSION AND FUTURE WORK

This paper proposed a fuzzy logic-driven adaptive semantic communication framework for Internet of Vehicles environments. By dynamically selecting transmission strategies according to channel conditions and vehicular dynamics, the proposed approach improves transmission quality while reducing unnecessary computational overhead. Experimental results demonstrated superior PSNR and accuracy performance compared with direct transmission methods and lower resource consumption than fully semantic communication schemes.

Future work will focus on:

- Real-world vehicular deployment
- Multi-user semantic communication
- Reinforcement learning-enhanced adaptive control
- Edge-assisted semantic communication
- Integration with 6G vehicular networks

ACKNOWLEDGMENT

The authors would like to thank XXX for supporting this research.